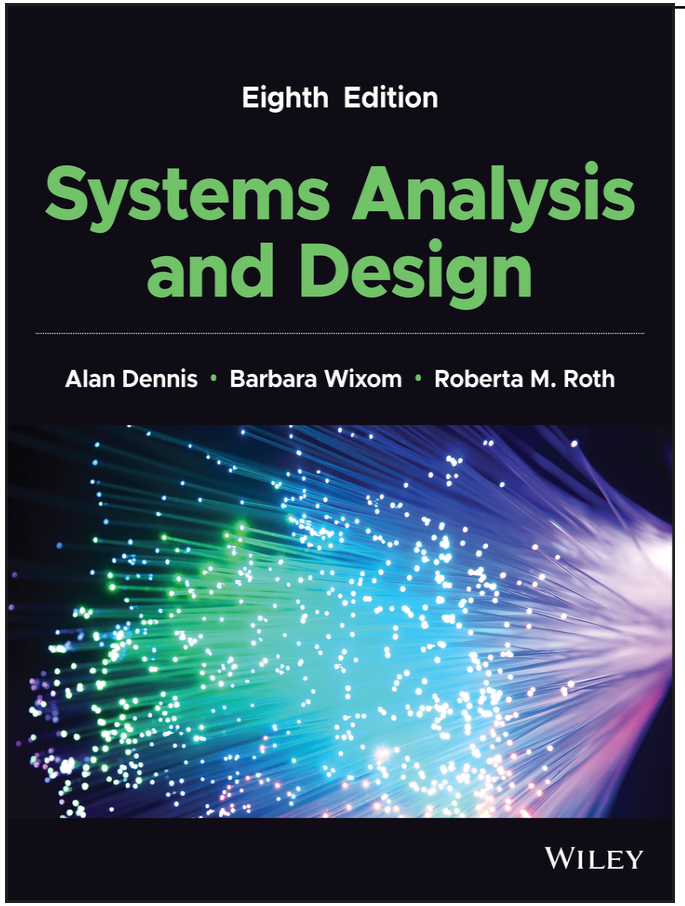
Software Development Lifecycles

Part 1



Systems Analysis and Design

Chapter 1

The Systems Analyst and Information Systems Development

OBJECTIVES

- The systems analyst.
- The Systems Development Life Cycle (SDLC).
- Information system project identification and initiation.
- Feasibility analysis.

Introduction

- The systems development life cycle (SDLC) is the process of determining how an information system (IS) can support business needs, designing the system, building it, and delivering it to users.
- The key person in the SDLC is the systems analyst, who analyzes the business situation, identifies the opportunities for improvements, and designs an IS to implement the improvements.

The systems analyst

- The systems analyst plays a key role in IS development projects.
- The systems analyst works closely with all project team members so that the team develops the right system in an effective way.
- Systems analysts must understand how to apply technology in order to solve problems.
- Systems analysts may serve as **change agents** who identify organizational improvement needed, design systems to implement those changes, and train and motivate others to use the systems.

Systems Analyst Skills

- Technical – Must understand the technical environment, technical foundation, and technical solution.
- Business – Must understand how IT can be applied to business situations.
- Analytical – Must be problem solvers.

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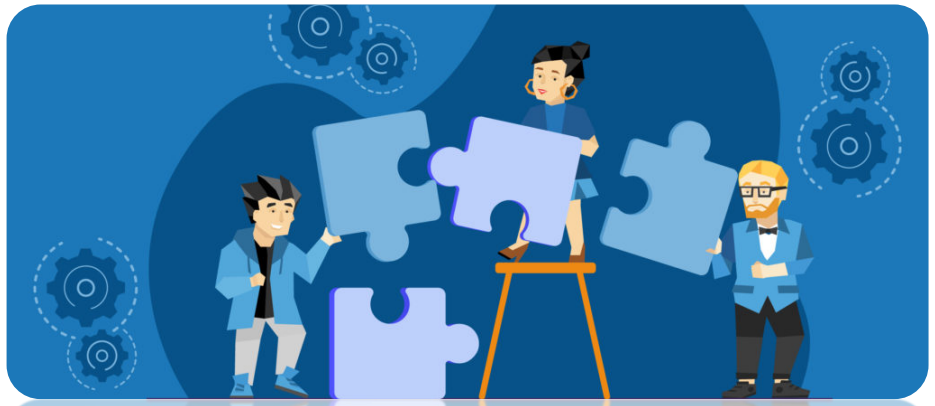
- Interpersonal – Need to communicate effectively.
- Management – Need to manage people and to manage pressure and risks.
- Ethical - Must deal fairly, honestly, and ethically with other project members, managers, and systems users.



Systems Analyst Roles

- **Business analyst** - Focuses on the IS issues surrounding the system.
- **Systems analyst** - Focuses on the business issues surrounding the system.
- **Infrastructure analyst** - Focuses on technical issues
- **Change management analyst** - Focuses on the people and management issues surrounding the system installation.
- **Project manager** - Ensures that the project is completed on time and within budget, and that the system delivers the expected vale to the organization.

Career Paths for Systems Analysts

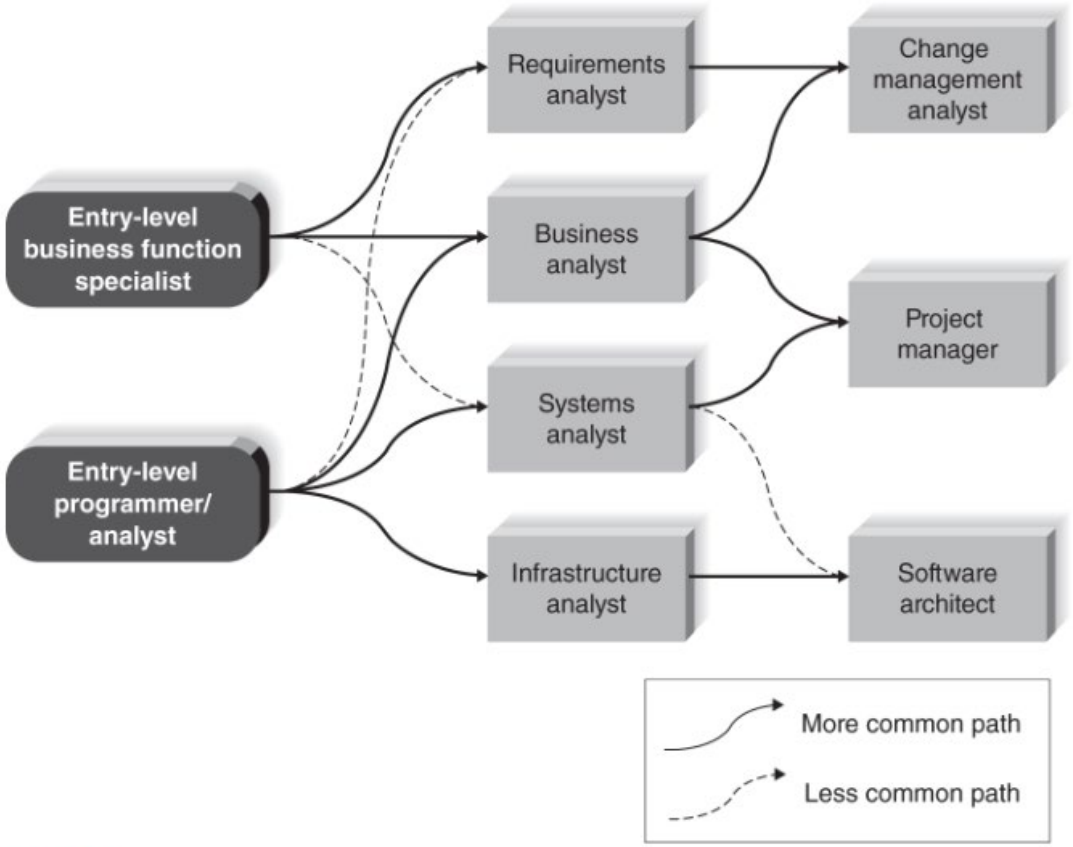


FIGURE 1-1 Career paths for system developers.

The Systems Development Life Cycle (SDLC)

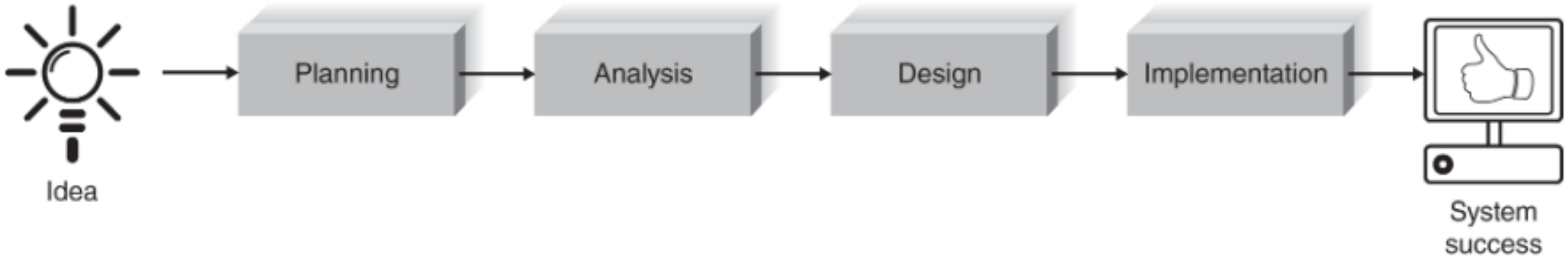


FIGURE 1-2 The systems development life cycle.

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- The **SDLC** is composed of four fundamental phases:
  - Planning
  - Analysis
  - Design
  - Implementation
- Each of the phases is composed of steps, which rely on techniques that produce *deliverables* (specific documents that explain various elements of the system).

Planning

- This phase is the fundamental process of understanding *why* an information system should be built, and determining *how* the project team will go about building it.

The planning phase has two steps:

1. During **project initiation**, the system's business value to the organization is identified (How will it lower costs or increase revenues?).
2. During **project management**, the project manager creates a work plan, staffs the project, and puts techniques in place to help the project team control and direct the project through the entire SDLC.

Analysis

- The analysis phase answers the questions of *who* will use the system, *what* the system will do, and *where* and *when* it will be used.
- During this phase the project team investigates any current system(s), identifies improvement opportunities, and develops a concept for the new system.

The analysis phase has three steps:

1. **Analysis strategy**: This is developed to guide the projects team's efforts. This includes a study of the current system and its problems, and envisioning ways to design a new system.

2. **Requirements gathering**: The analysis of this information leads to the development of a concept for a new system. This concept is used to build a set of analysis models.

3. **System proposal**: The proposal is presented to the project sponsor and other key individuals who decide whether the project should continue to move forward.

Design

- The design phase decides ***how*** the system will operate, in terms of the hardware, software, and network infrastructure; the user interface, forms, and reports that will be used; and the specific programs, databases, and files that will be needed.

The design phase has four steps:

1. **Design Strategy**: This clarifies whether the system will be developed by the company or outside the company.
2. **Architecture Design**: This describes the hardware, software, and network infrastructure that will be used.
3. **Database and File Specifications**: These documents define what and where the data will be stored.
4. **Program Design**: Defines what programs need to be written and what they will do.

Implementation

- During the implementation phase, the system is either developed or purchased (in the case of packaged software) and installed.
- This phase is usually the longest and most expensive part of the process.

The implementation phase has three steps:

1. **System Construction**: The system is built and tested to make sure it performs as designed.
2. **Installation**: The old system is turned off and the new one is turned on.
3. **Support Plan**: Includes a post-implementation review as well as a systematic way for identifying changes needed for the system.

Project identification and initiation

- A project is identified when someone in the organization identifies a ***business need*** to build a system.
- A need may surface when an organization identifies unique and competitive ways of using IT.
- To leverage the capabilities of ***emerging technologies*** such as cloud computing, RFID, Web 2.0

Business Process Management (BPM)

- Nowadays many new IS projects grow out of BPM.
- BPM is a methodology used by organizations to continuously improve end-to-end business processes.

BPM Process

- Defining and mapping the steps in a business process.
- Creating ways to improve on the steps in the process that add value
- Finding ways to eliminate or consolidate steps in the process that do not add value
- Creating and adjusting electronic workflows to match the improved process maps.

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- **Business process automation (BPA)** – technology components are used to complement or substitute manual process.
- **Business process improvement (BPI)** – creating new, re-designed processes to improve the workflows, and/or utilizing new technologies enabling new process structures.
- **Business process reengineering (BPR)** – changing the fundamental way in which the organization operate.

Project sponsor

- The project sponsor is a person (or group) who has an interest in the system's success
- The project sponsor will work throughout the SDLC to make sure that the project is moving in the right direction from the perspective of the business.
- The project sponsor serves as the primary point of contact for the project team.
- The size or scope of the project determines by the kind of sponsor that is involved.

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- The project sponsor has the insights needed to determine the business value that will be gained from the system.
- Tangible value can be quantified and measured easily (reduction in operating costs).
- An intangible value results from an intuitive belief that the system provides important, but hard-to-measure benefits to the organization.

System Request

- The document that describes the business reasons for building a system and the value that system is expected to provide.
- The project sponsor usually completes this form as part of a formal system selection process within the organization.

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- The business requirements of the project refer to the business capabilities that the system will need to have.
- The business value describes the benefits that the organization should expect from the system.
- Special issues are included on the document as a catchall category for other information that should be considered in assessing the project.

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- The completed system request is submitted to the approval committee for consideration.
- The committee reviews the system request and makes an initial determination of whether to investigate the proposed project or not.
- If so, the next step is to conduct a feasibility analysis.

Feasibility analysis

- Feasibility analysis guides the organization in determining whether to proceed with a project.
- Feasibility analysis also identifies the important risks associated with the project that must be managed if the project is approved.

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- As with the system request, each organization has its own process and format for the feasibility analysis, but most include techniques to assess three areas:
  - Technical feasibility
  - Economic feasibility
  - Organizational feasibility
- The results of evaluating these three feasibility factors are combined into a feasibility study deliverable that is submitted to the approval committee at the end of project initiation.

Technical Feasibility

- Technical feasibility is the extent to which the system can be successfully designed, developed, and installed by the IT group.
- It is, in essence, a technical risk analysis that strives to answer the question: “Can we build it?”

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- Risks can endanger the successful completion of a project. The following aspects should be considered:
  - Users' and analysts' should be familiar with the application.
  - Familiarity with the technology
  - Project size
  - Compatibility of the new system with the technology that already exists

Economic Feasibility

- Economic feasibility analysis is also called a cost-benefit analysis, that identifies the costs and benefits associated with the system.
- This attempts to answer the question: “Should we build the system?”

Cash Flow Analysis and Measures

- IT projects involve an initial investment that produces a steam of benefits over time, along with some on-going support costs.
- Cash flows, both inflows and outflows, are estimated over some future period.

Simple cash flow projection

		Year 0	Year 1	Year 2	Year 3	Total
	Total Benefits		45,000	50,000	57,000	152,000
	Total Costs	100,000	10,000	12,000	16,000	138,000
③	Net Benefits (Total Benefits –Total Costs)	(100,000)	35,000	38,000	41,000	14,000
④	Cumulative Net Cash Flow	(100,000)	(65,000)	(27,000)	14,000	

FIGURE 1-8 **Simple cash flow projection.**

Common methods for evaluating a project's worth

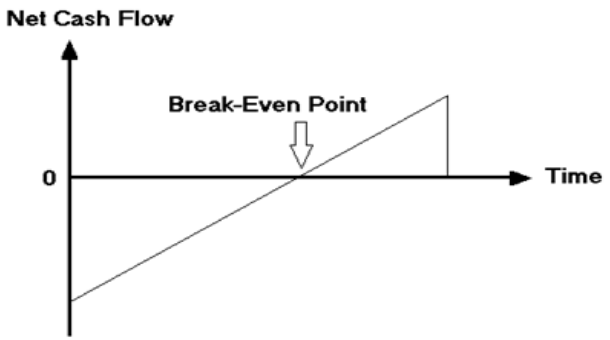
- Return on Investment (ROI)

$$\text{ROI} = \frac{\text{Total Benefits} - \text{Total Costs}}{\text{Total Costs}}$$

$$\text{ROI} = \frac{152{,}000 - 138{,}000}{138{,}000} = \frac{14{,}000}{138{,}000} = 10.14\%$$

ROI=(Total Benefits – Total Costs)/Total Costs

- Break-Even Point (BEP)



$$\text{BEP} = \text{Number of years of negative cash flow} + \frac{\text{That year's Net Cash Flow} - \text{That year's Cumulative Cash Flow}}{\text{That year's Net Cash Flow}}$$

Using the values in Figure 1-8, the BEP calculation is

$$\text{BEP} = 2 + \frac{41{,}000 - 14{,}000}{41{,}000} = 2 + \frac{28{,}000}{41{,}000} = 2.68 \text{ years}$$

Discounted cash flow technique

- Discounted case flows are used to compare the present value of all cash inflows and outflows for the project in the today's dollar terms.

$$\mathrm{PV} = \frac{\text{Cash flow amount}}{(1 + \text{Rate of return})^n}, \text{where } n \text{ is the year in which the cash flow occurs.}$$

- Net present value (NPV): the difference between the total PV of the benefits and the total PV of the costs.

Discounted cash flow projection

	Year 0	Year 1	Year 2	Year 3	Total
Total Benefits		45,000	50,000	57,000	
PV of Total Benefits		40,909	41,322	42,825	125,056
Total Costs	100,000	10,000	12,000	16,000	
PV of Total Costs	100,000	9,091	9,917	12,021	131,029

FIGURE 1-9 Discounted cash flow projection.

Steps to conduct an economic feasibility analysis

1. Identify Costs and Benefits
2. Assign Values to Costs and Benefits
3. Determine Cash Flow
4. Assess Project's Economic Value
   - ROI
   - BEP
   - NPV

Identify Costs and Benefits

- The costs and benefits and be broken down in to four categories:
  - Development costs
  - Operational costs
  - Tangible benefits
  - Intangibles

Development Costs	Operational Costs
Development team salaries Consultant fees Development training Hardware and software Vendor installation Office space and equipment Data conversion costs	Software upgrades Software licensing fees Hardware repair and upgrades Cloud storage fees Operational team salaries Communications charges User training
Tangible Benefits	**Intangible Benefits**
Increased sales Reductions in staff Reductions in inventory Reductions in IT costs Better supplier prices	Increased market share Increased brand recognition Higher-quality products Improved customer service Better supplier relations

FIGURE 1-11 Example of costs and benefits for economic feasibility.

Assign Values to C

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  assign specific dollar values to the

	2022	2023	2024	2025	2026	Total
Benefits						
Increased sales		500,000	530,000	561,800	595,508	2,187,308
Reduction in customer complaint calls[a]		70,000	70,000	70,000	70,000	280,000
Reduced inventory costs		68,000	68,000	68,000	68,000	272,000
Total Benefits[b]		**638,000**	**668,000**	**699,800**	**733,508**	**2,739,308**
Development Costs						
2 servers @ $125,000	250,000	0	0	0	0	250,000
Printer	100,000	0	0	0	0	100,000
Software licenses	34,825	0	0	0	0	34,825
Server software	10,945	0	0	0	0	10,945
Development labor	1,236,525	0	0	0	0	1,236,525
Total Development Costs	**1,632,295**	**0**	**0**	**0**	**0**	**1,632,295**
Operational Costs						
Hardware		50,000	50,000	50,000	50,000	200,000
Software		20,000	20,000	20,000	20,000	80,000
Operational labor		115,000	119,600	124,384	129,359	488,343
Total Operational Costs		**185,000**	**189,600**	**194,384**	**199,359**	**768,343**
Total Costs	**1,632,295**	**185,000**	**189,600**	**194,384**	**199,359**	**2,400,638**
Total Benefits − Total Costs	**(1,632,295)**	**453,000**	**478,400**	**505,416**	**534,149**	**338,670**
Cumulative Net Cash Flow	**(1,632,295)**	**(1,179,295)**	**(700,895)**	**(195,479)**	**338,670**	
Return on Investment (ROI)	**14.1% (338,670/2,400,638)**					
Break-Even Point	**3.37 years (3 years of negative cumulative cash flow + [534,149 − 338,670]/534,149 = 0.37)**					

[a] Customer service values are based on reduced costs of handling customer complaint phone calls.

[b] An important yet intangible benefit will be the ability to offer services that our competitors currently offer.

FIGURE 1-12 Cost–benefit analysis—simple cash flow method.

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Determine Cash Flow

- A formal cost-benefit analysis usually contains costs and benefits over a selected number or years to show cash flow over time.

  - Determine ROI

  - Determine BEP

  - Determine NPV

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	2022	2023	2024	2025	2026	Total
Benefits						
Increased sales		500,000	530,000	561,800	595,508	
Reduction in customer complaint calls[a]		70,000	70,000	70,000	70,000	
Reduced inventory costs		68,000	68,000	68,000	68,000	
Total Benefits[b]		**638,000**	**668,000**	**699,800**	**733,508**	
Present Value Total Benefits		**601,887**	**594,518**	**587,566**	**581,007**	**2,364,978**
Development Costs						
2 Servers @ $125,000	250,000	0	0	0	0	
Printer	100,000	0	0	0	0	
Software licenses	34,825	0	0	0	0	
Server software	10,945	0	0	0	0	
Development labor	1,236,525	0	0	0	0	
Total Development Costs	**1,632,295**	**0**	**0**	**0**	**0**	
Operational Costs						
Hardware		50,000	50,000	50,000	50,000	
Software		20,000	20,000	20,000	20,000	
Operational labor		115,000	119,600	124,384	129,359	
Total Operational Costs		**185,000**	**189,600**	**194,384**	**199,359**	
Total Costs	**1,632,295**	**185,000**	**189,600**	**194,384**	**199,359**	
Present Value Total Costs	**1,632,295**	**174,528**	**168,743**	**163,209**	**157,911**	**2,296,686**
NPV (PV Total Benefits – PV Total Costs)						**68,292**

[a]Customer service values are based on reduced costs of handling customer complaint phone calls.

[b]An important yet intangible benefit will be the ability to offer services that our competitors currently offer.

FIGURE 1-13 Cost–benefit analysis—discounted cash flow method.

Organizational Feasibility

- Organizational feasibility of the system is how well the system ultimately will be accepted by its users and incorporated into the ongoing operations of the organization.
- There are many organizational factors that can have an impact on the project, and seasoned developers know that organizational feasibility can be the most difficult feasibility dimension to assess.
- In essence, an organizational feasibility analysis is to answer the question "If we build it, will they come?"

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- One way to assess the organizational feasibility is to understand how well the goals of the project align with the business objectives and organizational strategies.
- A second way to assess the organizational feasibility is to conduct **stakeholder analysis.**
- A stakeholder is a person, group, or organization that can affect a new system
  - Project champion
  - System users
  - Organizational management
  - Other stakeholders

Summary

- The **Systems Analyst** is the key person in the development of information systems.
- **The Systems Development Lifecycle** consists of four stages: Planning, Analysis, Design, and Implementation.
- **Project Identification and Initiation** recognize a business need that can be satisfied through the use of information technology.
- **System Request** describes the business value for an information system.
- A **Feasibility Analysis** is used to provide more detail about the risks associated with the proposed system.